BRAC University
Department of Computer Science and Engineering

CSE 443: Bioinformatics-I

Final Exam: Fall 2025 Time: 1 hour 45 minutes Marks: 40

Name		ID		Section	

Answer any four of the following questions.

1. You are given the following distance matrix for four taxa $\{A, B, C, D\}$:

$$D = \begin{pmatrix} 0 & 4 & 7 & 7 \\ 4 & 0 & 7 & 7 \\ 7 & 7 & 0 & 4 \\ 7 & 7 & 4 & 0 \end{pmatrix}$$

   (a) Construct the Neighbor-Joining matrix D^* $(n = 4)$. [3]

$$D*(i,j) = (n-2)D(i,j) - \sum_k D(i,k) - \sum_k D(j,k),$$

   (b) Identify the pair of taxa selected as neighbors. Let u be the internal node joining the selected neighbors. Compute the branch length from each neighbor to u. [3]

   (c) Compute the distances from u to the remaining taxa and write down the reduced 3×3 distance matrix. [4]

2. Consider the following two DNA sequences: $S_1 : AGTCCGA, S_2 : TCCAG$. You are given the following scoring scheme: match score=+2, mismatch penalty=-1, gap penalty=-2. Using the Needleman–Wunsch algorithm, find the optimal global alignment(s) of the two sequences. Construct and clearly show the dynamic programming (DP) table. If multiple optimal alignments exist, report all of them. [10]

3. (a) 'Neural Networks are ensemble of logistic regression units' - explain this with an example. [5]

   (b) Explain how neural networks can be used to generate synthetic data and reduce dimensionality in practice. [5]

4. In a protein–protein interaction (PPI) network, nodes represent proteins and edges represent physical interactions. Such networks are commonly modeled as undirected, unweighted graphs.

   Consider the following graph with nodes A–E: A is connected to B and C; B is connected to A, C, and D; C is connected to A, B, and D; D is connected to B, C, and E; E is connected to D.

   Answer the following questions.

   (a) Draw the graph corresponding to the given description. Compute the PageRank score of each node after **two iteration**, assuming: initial PageRank is equal for all nodes, and since the graph is undirected, treat each edge as bidirectional. [10]

5. This problem explores the application of Hidden Markov Models (HMMs) to the task of gene finding in a newly sequenced genome segment. We define a simple HMM with two hidden states: 'G' (Gene region) and 'N' (Non-Gene/Intergenic region). The observable states are the four nucleotides: A, C, G, T. The core task in each sub-problem below is to decide which of the following four algorithms is most suitable: Viterbi Algorithm Forward Algorithm (or Forward-Backward when calculating probabilities) or Baum-Welch Algorithm.

A preliminary model has been defined with the following parameters.

Initial State Probabilities (π): $\pi_G = 0.1$ and $\pi_N = 0.9$

Transition Probabilities ($\mathbf{A}$): $P(\text{state}_t|\text{state}_{t-1})$

From / To	**G**	**N**
G	0.95	0.05
N	0.10	0.90

Emission Probabilities ($\mathbf{E}$): $P(\text{nucleotide}|\text{state})$

State	**A**	**C**	**G**	**T**
G	0.20	0.30	0.30	0.20
N	0.35	0.15	0.15	0.35

The short, observed DNA sequence is: $\mathbf{O}$ = A-G-C-G-T.

(a) You are tasked with determining the most probable location of the gene region within the observed sequence $\mathbf{O}$. This requires finding the single, highest-probability sequence of hidden states ($\mathbf{S}$) that could have produced $\mathbf{O}$ (e.g., N-G-G-G-N). Which of the three HMM algorithms (Viterbi, Forward, or Baum-Welch) is the correct tool for this decoding problem? Justify your choice. [3]

(b) You want to quantitatively assess how well the entire observed sequence $\mathbf{O}$ = A-G-C-G-T fits the current HMM parameters. This requires calculating the overall likelihood $P(\mathbf{O})$, which sums the probabilities over all possible hidden state paths. Which algorithm is required for this evaluation problem? Explain. [3]

(c) Assume a large, unlabeled training dataset of 100 new genomic sequences is now available. The initial parameters ($\mathbf{A}$ and $\mathbf{E}$) are suspected to be suboptimal for this genome. The task is to iteratively re-estimate these transition and emission probabilities to maximize the overall probability of generating the unlabeled training data. Which algorithm is required for this learning/training task when the hidden states (true gene boundaries) are unknown? [4]